Linear models

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Linear regression

Logistic regression

Model

We assume the output is a linear function of the input. The **linear regression** model is of the form

$$f(\boldsymbol{x}; \boldsymbol{w}, b) = w_1 x_1 + ... + w_D x_D + b = \boldsymbol{w}^\top \boldsymbol{x} + b$$

- $\boldsymbol{w}$: weights
- b : bias

b can be absorbed into $\boldsymbol{w}$ by defining $\boldsymbol{w} = [b, w_1, ..., w_D]$ and $\boldsymbol{x} = [1, x_1, ..., x_D]$, so that

$$f(\boldsymbol{x}; \boldsymbol{w}, b) = \boldsymbol{w}^\top \boldsymbol{x}$$

The general form of the linear regression model with all observations:

$$\hat{\boldsymbol{y}} = \boldsymbol{X}\boldsymbol{w} + b$$

- N : Number of observations
- D : number of features
- $\hat{\boldsymbol{y}} \in \mathbb{R}^N$: predictions
- $\boldsymbol{X} \in \mathbb{R}^{N \times D}$: input (design matrix)
- $\boldsymbol{w} \in \mathbb{R}^D$: weights
- $\boldsymbol{b} \in \mathbb{R}$: bias

When the bias b is absorbed

$$\hat{\boldsymbol{y}} = \boldsymbol{X}\boldsymbol{w}$$

Loss function - Least squares

Goal:

Find the parameters $\boldsymbol{w}$ that minimize the **residual sum of squares** (loss)

$$\text{RSS}(\boldsymbol{w}) = \frac{1}{2}\sum_{i=1}^{N}(y_i - f(\boldsymbol{x}_i))^2 = \frac{1}{2}\sum_{i=1}^{N}(y_i - \boldsymbol{w}^\top \boldsymbol{x}_i)^2$$

We can minimize it analytically or iteratively using **gradient descent**.

Probabilistic Interpretation

The targets and inputs are related as follows

$$\boldsymbol{y} = \boldsymbol{w}^\top \boldsymbol{x} + \epsilon$$

where ϵ is the residual error between the predictions and the true response (unmodeled effects/random noise). We assume ϵ has a Gaussian distribution $\epsilon \sim \mathcal{N}(0, \sigma^2)$.

$$p(y|\boldsymbol{x}; \boldsymbol{\theta}) = \mathcal{N}(y; \boldsymbol{w}^\top \boldsymbol{x}, \sigma^2)$$

where $\boldsymbol{\theta} = (\boldsymbol{w}, \sigma^2)$.
We estimate the parameters using **Maximum Likelihood Estimation**. We want the parameters that maximizes the likelihood $\prod_{i=1}^{N} p(y_i|\boldsymbol{x}_i; \boldsymbol{\theta})$. It is easier to minimize the **Negative log likelihood**

$$\text{NLL}(\boldsymbol{\theta}) = -\sum_{i=1}^{N} \log p(y_i|\boldsymbol{x}_i; \boldsymbol{\theta})$$

It can be shown that minimizing the NLL is equivalent to minimizing the RSS.

Ordinary Least Squares

Our loss function is

$$J(\boldsymbol{w}) = \text{RSS}(\boldsymbol{w}) = \frac{1}{2}\sum_{i=1}^{N}(y_i - \boldsymbol{w}^\top \boldsymbol{x}_i)^2 = \frac{1}{2}||\boldsymbol{X}\boldsymbol{w} - \boldsymbol{y}||_2^2 = \frac{1}{2}(\boldsymbol{X}\boldsymbol{w} - \boldsymbol{y})^\top(\boldsymbol{X}\boldsymbol{w} - \boldsymbol{y})$$

The gradient is given by

$$\nabla_{\boldsymbol{w}}\text{RSS}(\boldsymbol{w}) = \boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{w} - \boldsymbol{X}^\top\boldsymbol{y}$$

Setting the gradient to zero

$$\boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{w} = \boldsymbol{X}^\top\boldsymbol{y}$$

called the **normal equations**.
The solution $\hat{\boldsymbol{w}}$ called the **ordinary least squares** solution is given by

$$\hat{\boldsymbol{w}} = (\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top\boldsymbol{y}$$

Is it a unique global minimum ?

We check if the Hessian is positive definite. It is given by

$$\mathrm{H}(\boldsymbol{x}) = \frac{\partial}{\partial \boldsymbol{w}} \mathrm{RSS}(\boldsymbol{w}) = \boldsymbol{X}^\top \boldsymbol{X}$$

If the columns of $\boldsymbol{x}$ are linearly independent, then H is positive definite and $\hat{\boldsymbol{w}}$ is a unique global minimum.

Numerical issues

The inverse should not be computed directly. $\boldsymbol{X}^\top \boldsymbol{X}$ can be singular or ill-conditioned. There are alternatives:

- SVD
- QR decomposition

Explore further

- Polynomial regression (other basis expansions)
- Weighted linear regression
- Bayesian linear regression

Outline

Linear regression
- Ridge regression
- Lasso regression

Logistic regression

Model

To avoid overfitting, a common strategy is to constrain the weights to be smaller. Weights that become too large in magnitude are penalized. This is called **weight decay** or L^2 **regularization**.
The loss function is updated as follows

$$J(\boldsymbol{w}) = \text{RSS}(\boldsymbol{w}) + \lambda ||\boldsymbol{w}||_2^2$$

where λ controls the strength of the regularizer

Probabilistic interpretation

Instead of MLE, we use MAP estimation with a zero-mean Gaussian prior on the weights $p(\boldsymbol{w}) = \prod_{i=1}^{D} \mathcal{N}(w_i; 0, \tau^2)$. The MAP estimate corresponds to minimizing

$$J(\boldsymbol{w}) = \frac{1}{2\sigma^2}||\boldsymbol{X}\boldsymbol{w} - \boldsymbol{y}||_2^2 + \frac{1}{2\tau^2}||\boldsymbol{w}||_2^2$$

where $\lambda = \frac{\sigma^2}{\tau^2}$

$$J(\boldsymbol{w}) = ||\boldsymbol{X}\boldsymbol{w} - \boldsymbol{y}||_2^2 + \lambda||\boldsymbol{w}||_2^2 = \text{RSS}(\boldsymbol{w}) + \lambda||\boldsymbol{w}||_2^2$$

Solving the MAP estimate

The gradient is given by

$$\nabla_{\boldsymbol{w}} J(\boldsymbol{w}) = 2(\boldsymbol{X}^\top \boldsymbol{X}\boldsymbol{w} - \boldsymbol{X}^\top \boldsymbol{y} + \lambda \boldsymbol{w})$$

we set the gradient to zero to obtain

$$\hat{\boldsymbol{w}} = (\boldsymbol{X}^\top \boldsymbol{X} + \lambda \boldsymbol{I}_D)^{-1} \boldsymbol{X}^\top \boldsymbol{y}$$

Numerical issues

As in standard linear regression, computing directly the inverse is slow and could be numerically unstable. There are other ways.

- Convert it to standard least squares then use QR decomposition
- SVD

Outline

Linear regression
- Ridge regression
- Lasso regression

Logistic regression

Model

We want the parameters to no just be small as in ridge regression, but to be exactly zero. l_1 norm is used for the penalty. This way, $\hat{\boldsymbol{w}}$ is sparse. This approach is called lasso or least absolute shrinkage and selection operator.
The loss function becomes

$$J(\boldsymbol{w}) = \text{RSS}(\boldsymbol{w}) + \lambda ||\boldsymbol{w}||_1$$

where λ controls the strength of the regularizer

Probabilistic interpretation

We use MAP estimation with a Laplace prior on the weights $p(\boldsymbol{w}; \lambda) = \prod_{i=1}^{D} Lap(w_i; 0, 1/\lambda)$. The MAP estimate corresponds to minimizing the previous objective function

$$J(\boldsymbol{w}) = ||\boldsymbol{X}\boldsymbol{w} - \boldsymbol{y}||_2^2 + \lambda||\boldsymbol{w}||_1$$

Optimization

There is no analytical solution as in linear and ridge regression. However, the objective function is still convex.

Explore further

Solve Lasso using:

- LARS (least angle regression and shrinkage)
- Gradient descent
- Coordinate descent

Combine lasso and ridge to obtain elastic net whose objective is:

$$J(\boldsymbol{w}) = ||\boldsymbol{X}\boldsymbol{w} - \boldsymbol{y}||_2^2 + \lambda_2||\boldsymbol{w}||_2^2 + \lambda_1||\boldsymbol{w}||_1$$

Linear regression

Logistic regression

Model

It is a classification model $p(y|\boldsymbol{x};\boldsymbol{\theta})$.

- $\boldsymbol{x} \in \mathbb{R}^D$: input
- $y \in \{1, ..., C\}$: class label
- $\boldsymbol{\theta}$: parameters

If $C = 2$, it is called **binary logistic regression** and if $C > 2$, it is known as **multiclass logistic regression**.

Binary logistic regression

Since we want to predict $y \in 0, 1$ given some inputs $\boldsymbol{x}$, the model is of the form

$$p(y|\boldsymbol{x}; \boldsymbol{\theta}) = \text{Ber}(y; f(\boldsymbol{x}; \boldsymbol{\theta}))$$

where $f(\boldsymbol{x}; \boldsymbol{\theta})$ is a function giving the parameter of the distribution hence must satisfy $0 \leq f(\boldsymbol{x}; \boldsymbol{\theta}) \leq 1$. To allow f to be any function, we use:

$$p(y|\boldsymbol{x}; \boldsymbol{\theta}) = \text{Ber}(y; \sigma(f(\boldsymbol{x}; \boldsymbol{\theta})))$$

where σ is the **sigmoid** (S-shaped) or **logistic** function:

$$\begin{aligned} \sigma :& \mathbb{R} \to [0, 1] \\ & z \mapsto \sigma(z) = \frac{1}{1 + e^{-z}} \end{aligned}$$

z is called the logit or the pre-activation.

For logistic regression, we choose a linear function $f(\boldsymbol{x};\boldsymbol{\theta}) = \boldsymbol{w}^\top \boldsymbol{x} + b$. The model has the form

$$p(y|\boldsymbol{x};\boldsymbol{\theta}) = \text{Ber}(y;\sigma(\boldsymbol{w}^\top \boldsymbol{x} + b))$$

This means

$$p(y = 1|\boldsymbol{x};\boldsymbol{\theta}) = \sigma(\boldsymbol{w}^\top \boldsymbol{x} + b)) = \frac{1}{1 + e^{-(\boldsymbol{w}^\top \boldsymbol{x} + b)}}$$

Decision boundary

During prediction, we have

$$\hat{y} = \begin{cases} 1 & \text{if } p(y = 1|\boldsymbol{x}; \boldsymbol{\theta}) > 0.5 \\ 0 & \text{if } p(y = 1|\boldsymbol{x}; \boldsymbol{\theta}) < 0.5 \end{cases}$$

which is the same as

$$\hat{y} = \begin{cases} 1 & \text{if } \boldsymbol{w}^\top \boldsymbol{x} + b > 0 \\ 0 & \text{if } \boldsymbol{w}^\top \boldsymbol{x} + b < 0 \end{cases}$$

The **decision boundary** is $\boldsymbol{w}^\top \boldsymbol{x} + b = 0$. It is a linear hyperplane with normal vector $\boldsymbol{w}$ and an offset b from the origin. It separates the space into 2 half-spaces.
The data is said to be **linearly separable** when the examples can be perfectly separated by the linear hyperplane.

Maximum likelihood estimation

We note $\mu_i = \sigma(z_i) = \sigma(\boldsymbol{w}^\top \boldsymbol{x}_i)$. The negative log likelihood is

$$\begin{aligned}
\text{NLL}(\boldsymbol{w}) &= -\sum_{i=1}^{N} \log p(y_i|\boldsymbol{x}_i;\boldsymbol{\theta}) = -\sum_{i=1}^{N} \log Ber(y;\mu_i) \\
&= -\sum_{i=1}^{N} \log[\mu_i^{y_i}(1-\mu_i)^{1-y_i}] \\
&= -\sum_{i=1}^{N} [y_i \log(\mu_i) + (1-y_i)\log(1-\mu_i)] \\
&= \sum_{i=1}^{N} \mathbb{H}(y_i, \mu_i)
\end{aligned}$$

where $\mathbb{H}$ is the **binary cross entropy**. This objective is convex and can be minimized using gradient-based methods.

Multinomial logistic regression

It is a classification model of the form

$$p(y|\boldsymbol{x};\boldsymbol{\theta}) = \text{Cat}(y; f(\boldsymbol{x};\boldsymbol{\theta}))$$

We note $\mu = f(\boldsymbol{x};\boldsymbol{\theta})$ (here $f : \mathbb{R}^D \to \mathbb{R}^C$) . It must satisfy $0 \leq \mu_i \leq 1$ and $\sum_{i=1}^{C} \mu_i = 1$. To allow f to be any function, we pass it to the **softmax** function

$$\begin{aligned} \mathcal{S} : \mathbb{R}^C &\to [0,1]^C \\ z &\mapsto \mathcal{S}(z) = \left[\frac{e^{z_1}}{\sum_{i=1}^{C} e^{z_i}}, ..., \frac{e^{z_C}}{\sum_{i=1}^{C} e^{z_i}} \right] \end{aligned}$$

You might want to use the log-sum-exp trick to avoid numerical overflow when computing the softmax.

We use a linear function $f(\boldsymbol{x}; \boldsymbol{\theta}) = \boldsymbol{W}\boldsymbol{x} + \boldsymbol{b}$ where $\boldsymbol{W}$ is a $C \times D$ matrix and $\boldsymbol{b}$ is a C dimensional vector. The model is of the form

$$p(y|\boldsymbol{x}; \boldsymbol{\theta}) = \text{Cat}(y; \mathcal{S}(\boldsymbol{W}\boldsymbol{x} + \boldsymbol{b}))$$

If we note $\boldsymbol{z} = \boldsymbol{W}\boldsymbol{x} + \boldsymbol{b}$ the C dimensional vector of logits, we have

$$p(y = c|\boldsymbol{x}; \boldsymbol{\theta}) = \frac{e^{z_c}}{\sum_{i=1}^{C} e^{z_i}}$$

Maximum likelihood estimation

The negative log likelihood is

$$\begin{aligned} \text{NLL}(\boldsymbol{w}) &= -\log \prod_{i=1}^{N} \prod_{c=1}^{C} \mu_{ic}^{y_{ic}} \\ &= -\sum_{i=1}^{N} \sum_{i=c}^{C} y_{ic} \log \mu_{ic} \\ &= \sum_{i=1}^{N} \mathbb{H}(y_i, \mu_i) \end{aligned}$$

where $\mu_{ic} = p(y_i = c|x_i; \boldsymbol{\theta}) = (\mathcal{S}(\boldsymbol{W}\boldsymbol{x}_i + \boldsymbol{b}))_c$ and $y_{ic} = \mathbb{I}(y_i = c)$. This objective is also convex and can be minimized using gradient descent.

Do you want more ?

Of course you do !

Explore further

- Robust logistic regression
- Bayesian logistic regression
- Multilabel classification
- Hierarchical classification